

Bumjun Kim

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Education

Yonsei University

Ph.D. in Artificial Intelligence

◦ GPA: 4.24/4.30

Mar 2025 – Expected Feb 2030

Seoul, Korea

Hongik University

B.S. in Computer Engineering

◦ GPA: 3.89/4.50

Mar 2018 – Feb 2025

Seoul, Korea

Publications

Peer-Reviewed Publications (* equal contribution, [†] corresponding author)

1. DAPD: Dependency-Aware Parallel Decoding via Attention for Diffusion LLMs

Bumjun Kim^{*}, Dongjae Jeon^{*}, Moongyu Jeon^{*}, Albert No[†]

ICML 2026

2. Rainbow Padding: Mitigating Early Termination in Instruction-Tuned Diffusion LLMs

Bumjun Kim^{*}, Dongjae Jeon^{*}, Dueun Kim^{*}, Wonje Jeung, Albert No[†]

ICLR 2026

3. Memorization In Stable Diffusion Is Unexpectedly Driven by CLIP Embeddings

Bumjun Kim, Albert No[†]

CVPR 2026 Findings

4. Same Trajectory, Contradictory Rewards (ROBORMBENCH): Paraphrase Fragility in Vision Language Reward Models

Wonje Jeung, Sangyeon Yoon, Hyesoo Hong, Yoonjun Cho, Dongjae Jeon, Bumjun Kim, Jean Oh[†], Youngjae Yu[†], and Albert No[†]

EMNLP 2026 Main

Workshop Papers

1. Single-Step Initialization for Exploratory Parallel Rollouts in Diffusion LLMs

Dongjae Jeon^{*}, Bumjun Kim^{*}, Mingyu Kim[†], and Albert No[†]

SPIGM Workshop at ICML 2026

2. A Theoretical Analysis of Why Masked Diffusion Models Mitigate the Reversal Curse

Moongyu Jeon^{*}, Sangwoo Shin^{*}, Bumjun Kim, Kyelim Lee, Albert No[†]

FoGen Workshop at ICML 2026

Research Experience

Ph.D. Researcher

Mar 2025 – Present

Artificial Intelligence & Information Systems Laboratory (AI-ISL)

Seoul, Korea

Advisor: Prof. Albert No

I study the underlying mechanisms that shape the capabilities and limitations of diffusion language models and use these insights to develop more reliable and efficient generation methods.

- **Parallel decoding:** Formulated diffusion-LM parallel decoding as a graph coloring problem using attention-derived dependency graphs, and developed DAPD, a training-free method inspired by Welsh–Powell, to improve decoding efficiency. (*ICML 2026*)
- **Diffusion-LM fine-tuning:** Developed Rainbow Padding to mitigate premature termination by replacing repeated EOS padding with a repeating cycle of distinct padding tokens, improving length robustness with lightweight LoRA fine-tuning. (*ICLR 2026*)
- **RL post-training:** Developed a one-step initialization method that diversifies diffusion-LM rollouts by unmasking one randomly selected position before standard decoding, improving exploration during RL post-training. (*ICML 2026 SPIGM Workshop*)

Undergraduate Researcher

Jul 2023 – Feb 2025

Artificial Intelligence & Information Systems Laboratory (AI-ISL)

Seoul, Korea

Advisor: Prof. Albert No

- Analyzed how duplicated CLIP padding embeddings amplify memorization in Stable Diffusion and developed inference-time masking interventions to reduce memorization without retraining. (*CVPR 2026 Findings*)

Honors & Awards

Science and Technology Scholarship (Undergraduate)

Sep 2022 – Feb 2025

Hyundai Motor Chung Mong-Koo Foundation

Received full tuition support plus a stipend of approximately USD 1,500 per semester.

Google Solution Challenge

May 2024

Top 100 Finalist

Biohealth Data Competition – Dentistry Track (Hongik University)

Dec 2023

Encouragement Award

Yongin City SW/AI Hackathon

Oct 2023

Grand Prize (Director's Award, Yongin Industry Promotion Agency)

Selected Activities

21st Cohort, Data Analysis, BOAZ

Jul 2023 – Jul 2024

Intercollegiate Big Data Organization

Member, GDG on Campus Hongik Univ.

Sep 2023 – Feb 2025

Hongik University